MIL-C-12927B (MU)
12 NOVEMBER 1963
SUPERSEDING
MIL-C-12927A
30 OCTOBER 1962

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M2E1 PARTS FOR, LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the parts for, loading, assembling and packing for one type of ignition cartridge designated as Cartridge, Ignition, M2E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 —Ammunition and Special Weapons; General Specification for
MIL-P-223 —Powder, Black
MIL-I-45208 —Inspection Requirements, General Specification for
MIL-I-45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105D—Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 —Quality Assurance Terms and Definitions
MIL-STD-129 —Marking for Shipment and Storage
MIL-STD-1235 —Single and Multi Level Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8863425—Cartridge, Ignition, M2E1 Assembly

PUBLICATIONS

ORDNANCE CORPS

IEL-8863425—Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

FSC: 1315

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CODE OF FEDERAL REGULATIONS
49 CFR 71–90—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite "49 CFR 71–90 (latest revisions)."

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Cartridge assembly. The assembly shall comply with all requirements specified on Drawing (dwg.) 8863425 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder. The moisture content of the black powder at the loading station, at the time of loading, shall not exceed 0.50 percent when determined as specified in 4.3.1.

3.4 Primer Functioning Sensitivity.

3.4.1 *Nonfunctioning.* The primer and body assembly shall not function under an applied energy value of 4 inch ounces minimum (min.) when tested as specified in 4.3.2.1.

3.4.2 *Functioning.* The primer and body assembly shall function under an applied energy value of 1920 inch ounces maximum (max.) when tested as specified in 4.3.2.2.

3.5 Waterproofness. The loaded cartridges shall function after being completely submerged in water for 3 hours (see 4.3.3).

3.6 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, blisters and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

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(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analyses on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 **Inspection provisions.**

4.2.1 *Lot formation.* A lot shall consist of cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL–STD–105D (or Standard MIL–STD–1235, when applicable). Each lot shall contain:
(a) Primers of one lot interfix number from one manufacturer.
(b) Black powder of one lot interfix number from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105D except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Body assembly (see dwg. 8863423 covering a detail of 8863424).

Categories	Defects	Method of inspection	Code No. (see 6.2)
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Diameter of flange (max.)	Gage	01001
102.	Outside diameter of body, max.	Gage	01002
103.	Knurl not 360 degrees	Visual	01003
104.	Head loose	Visual-Manual	01004
105.	Wad missing or improperly seated	Visual	01005
Minor:	AQL 1.0 percent		
201.	Marking misleading or unidentifiable	Visual	01006
202.	Cut or blister on inside of body	Visual	01007

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4.2.2.2 Primer and body assembly (see dwg. 8863424 covering a detail of 8863425).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Primer above flush	Straightedge	02001
Major:	AQL 0.25 percent		
101.	Primer flash hole not completely covered by lacquer	Visual	02002
102.	Primer loose or missing	Visual-Manual	02003
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	02004

4.2.2.3 Crimp cap (see dwg. 8863421 covering a detail of 8863425).

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.50 percent		
201.	Length, max.	Gage	03001
202.	Inside diameter	Gage	03002
203.	Wall thickness, min.	Gage	03003
204.	Evidence of poor workmanship (see 3.6)	Visual	03004

4.2.2.4 Cartridge assembly prior to crimping (see dwg. 8863425).

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Weight of charge, out of tolerance	Balance	04001
Minor:	AQL 0.40 percent		
201.	Foreign matter in charge	Visual	04002

4.2.2.5 Cartridge assembly prior to assembling crimp cap (see dwg. 8863425).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Wadding slanted or damaged causing power leakage	Visual	05001
Major: None defined.			
Minor:	AQL 1.50 percent		
201.	Crimp roll not 360 degrees	Visual	05002
202.	Lacquer missing or improperly applied	Visual	05003
203.	Wadding slanted or damaged with no power leakage	Visual	05004
204.	Marking on outer wad misleading or unidentifiable	Visual	05005
205.	Evidence of poor workmanship (see 3.6)	Visual	05006

4.2.2.6 Ignition cartridge assembly (see dwg. 8863425).

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Outside diameter at crimp cap max.	Gage	06001
102.	Overall length, max.	Gage	06002
103.	Lacquer missing from end of primer or crimp cap	Visual	06003

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 1.50 percent		
201.	Crimp cap loose	Manual	06004
202.	Crimp cap knurl not 360 degrees	Visual	06005
203.	Marking misleading or unidentifiable	Visual	06006
204.	Evidence of poor workmanship (see 3.6)	Visual	06007

4.2.3 *Testing.*

4.2.3.1 *Moisture content of black powder (see 3.3)—major defect, Code No. 04003.* The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test for verification at least one completed cartridge assembly from each eight hours production. Composite samples shall not be used in determining compliance. The black powder shall be tested as specified in 4.3.1 and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.2 *Sensitivity.*

4.2.3.2.1 *Nonfunctioning (see 3.4.1)—Critical defect, Code No. 02005.* The sampling plan for this test for each lot produced shall be in accordance with Standard MIL–STD–105D, table II–A, Code letter L and AQL of 0.065 percent. The test shall be performed as specified in 4.3.2.1.

4.2.3.2.2 Functioning (see 3.4.2) major defect, Code No. 02006.

4.2.3.2.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table III–A, Code letter M and AQL of 0.25 percent. The test shall be performed as specified in 4.3.2.2.

4.2.3.2.2.2 After three consecutive lots have met the criteria of 4.2.3.2.2.1, the sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table III–A, Code letter L and AQL of 0.25 percent. The test shall be performed as specified in 4.3.2.2.

4.2.3.3 Waterproofness (see 3.5) major defect, Code No. 06008.

4.2.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table II–A, Code letter J and AQL of 0.65 percent. The test shall be performed as specified in 4.3.3.

4.2.3.3.2 After three consecutive lots have met the criteria of 4.2.3.3.1, the sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table II–A, Code letter G and AQL of 0.40 percent. The test shall be performed as specified in 4.3.3.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Lists number IEL–8863425 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor

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shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Moisture content of black powder.* The moisture content of the black powder shall be determined in accordance with the procedure specified in MIL–P–223.

4.3.2 *Primer Sensitivity.*

4.3.2.1 *Nonfunctioning test.* The primer and body assembly shall be assembled to a test fixture (see 4.2.4) so that the firing pin will strike the sensitive end of the primer. A 1.94 plus or minus .02 ounce weight shall be dropped on the firing pin from a height of (2) inches (min.). The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight.

4.3.2.2 *Functioning test.* The procedure used shall be the same as in 4.3.2.1 except that a weight of 10—0.2 pounds shall be dropped on the firing pin from a height of 12 inches (max.).

4.3.3 *Waterproofness.* The ignition cartridge assemblies shall be completely immersed in water at a temperature of 70 plus or minus 5 degrees Fahrenheit (F.) for a 3-hour period. After the immersion period the cartridges shall be removed from the water, wiped dry, assembled in inspection equipment in accordance with 4.2.4 and fired by an impact equal to that specified in 4.3.2.2. Any cartridge that fails to fire shall be classed as defective.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 *Level C.* Unless otherwise specified the cartridge assemblies shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. The packing container shall comply with the requirements of Code of Federal Regulations 49 CFR 71–90 and shall be sufficiently rigid to withstand transportation and permit storing in tiers at least 10 feet high without damage to the container or contents.

5.2 **Labeling and marking.** The labeling and marking of the containers shall be in accordance with Standard MIL–STD–129 and Code of Federal Regulations 49 CFR 71–90.

5.3 **Data cards.** Data card information shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 **Ordering data.** Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 **Inspection equipment.** The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor's design responsibility on contractor format.
OD —Ordnance design.
ODMU —Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs

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is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Check test for deterioration (see 6.5). If the total elapsed time between original acceptance of the ignition cartridge lot and the assembly of that lot into the complete round exceeds two years, or if the ignition cartridges have been subjected to adverse conditions for any period of time since previous tests, the ignition cartridges shall be subjected to, and must satisfactorily comply with the following tests immediately before the ignition cartridges are assembled into the complete round.

6.4.1 *Ignition cartridge assembly (after removal of black powder).*

6.4.1.1 *Nonfunctioning.* The sampling plan for this test shall be as specified in 4.2.3.2.1. The test shall be performed as specified in 4.3.2.1.

6.4.1.2 *Functioning.* The sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table II–A, Code letter L and AQL of 0.65 percent. The test shall be performed as specified in 4.3.2.2.

6.4.2 *Moisture content of black powder.* Twenty-five samples of black powder shall be selected from the ignition cartridges disassembled for the test specified in 6.4.1. The moisture content shall not exceed 0.50 percent when determined as specified in 4.3.1.

6.5 **Cost of check test.** The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the ignition cartridges or to the contractor assembling the ignition cartridges in the complete rounds and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the primer lots were under the jurisdiction of either contractor.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. 1315–A354

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1426 APR 62 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-DC7
Dover, New Jersey

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